

René MOURAMBOU DIYEMA

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#) | [Medium](#) | [LeetCode](#) | [HackerRank](#) | Paris, France

I'm a **Quantitative Analyst** with an **MSc in Probability, Statistics and Finance** from Gustave Eiffel University. I **develop, calibrate and validate stochastic models for equity and fixed-income derivative pricing and risk**, turning market data into prices, hedges and risk measures through stochastic calculus, Monte Carlo methods and statistical inference. I am seeking a **Quantitative Finance internship starting in April 2027**.

EDUCATION

MSc Probability, Statistics and Finance — <i>Gustave Eiffel University, France</i>	Sept 2026 – Sept 2027
<i>Courses: High-dimensional statistics, simulations and copulas, empirical estimation and extreme values, deterministic and stochastic approximation, stochastic calculus, fractional models in finance, statistical learning, derivatives, high-frequency data and order books, interest rates and credit</i>	
MSc Artificial Intelligence — <i>Paris-Saclay University, France</i>	Sept 2025 – Sept 2026
<i>Courses: Probabilistic generative models, signal processing, frugal AI, theory and advanced machine learning, advanced optimization</i>	
MSc Mathematics, Vision, Learning — <i>ENS Paris-Saclay, France</i>	Sept 2024 – Sept 2025
<i>Courses: Stochastic calculus, statistical learning with extreme values, time series analysis, computational statistics, convex optimization</i>	
MSc Applied Mathematics — <i>ESIEE Institute of Technology, France (gap year 2023–2024)</i>	Sept 2022 – Sept 2024
<i>Courses: Probability for finance, discrete Markov chains, stochastic calculus, financial time series, jump models, risk measures, financial markets, Fixed Income, Equities, Portfolio Management, Interest Rates</i>	
BSc Applied Mathematics — <i>Sup'Management Engineering School, Gabon</i>	Sept 2017 – Sept 2019
<i>Courses: Linear algebra, calculus, probability and statistics, Markov processes, optimization</i>	

PROFESSIONAL EXPERIENCE

Machine Learning Researcher Full-Time Internship Nokia, France	Apr 2026 – Sep 2026
- Structured large-scale time-series signals for statistical estimation, controlling data quality and consistency across sources. - Characterised the series as stochastic processes, estimating temporal dependence, stationarity and distributional properties. - Applied statistical signal processing (filtering, spectral analysis) to isolate the components used for model estimation. - Specified and estimated temporal models (CNN, ResNet, Transformer) under supervised, semi-supervised and self-supervised regimes. - Benchmarked out-of-sample performance through hypothesis testing against reliability and generalization criteria. - Ran large-scale numerical experiments on GPU, comparing specifications and tracking estimation results across trials. - Documented model assumptions and estimation procedures, and presented results to business and engineering teams to guide decisions. Technologies: Python, NumPy, SciPy, PyTorch, Time Series, Stochastic Processes, Signal Processing, Statistical Inference, Hypothesis Testing, CUDA	
Machine Learning Researcher Full-Time Internship SEGULA Technologies, France	Apr 2025 – Sep 2025
- Prepared high-dimensional datasets and constructed informative features to support forecasting. - Developed reduced-order models for multi-physics systems governed by nonlinear PDEs. - Implemented Galerkin projection with error control and numerical convergence for finite- and infinite-order dynamics. - Modelled system dynamics with Autoencoders, LSTM and GNNs to produce multi-step forecasts. - Quantified forecast uncertainty and stability through residual analysis, refining specifications to reduce prediction error. - Tracked forecast performance over time to detect drift and recalibrate model parameters. Technologies: Python, NumPy, SciPy, PyTorch, Nonlinear PDEs, Galerkin, Reduced-Order Modeling, Forecasting, LSTM, GNNs, CUDA	
Machine Learning Researcher Full-Time Internship Evolukid, France	Jan 2024 – Jun 2024
- Structured and annotated multimodal datasets to support probabilistic evaluation. - Estimated probabilistic models for classification and detection, combining text and image signals for decision-making. - Evaluated decision rules via controlled A/B experiments in a real-time decision loop, measuring latency and decision quality. Technologies: Python, PyTorch, Pandas, Probabilistic Models, A/B Testing, Statistical Evaluation, Decision Models	

PROJECTS

Exotic Derivatives: Autocallable Pricing, Hedging and Model Risk	Sept 2026 – Oct 2026
- Built an arbitrage-free implied volatility surface from S&P 500 options and derived Dupire local volatility. - Built Heston and local-stochastic volatility models with a leverage function on the same surface. - Valued a 3-year autocallable and a barrier reverse convertible by Monte Carlo under each model for model risk. - Computed Greeks by bumping, simulated weekly delta hedging and decomposed profit and loss by Greek. Technologies: Python, QuantLib, Monte Carlo, Implied and Local Volatility, Heston, Local-Stochastic Volatility, Autocallables, Delta Hedging	
Rough Volatility: Calibration and Monte Carlo Pricing under Rough Bergomi	Apr 2026 – May 2026
- Estimated the Hurst exponent from realized volatility series to characterise rough volatility dynamics. - Simulated fractional Brownian motion with the hybrid scheme to implement the rough Bergomi model. - Priced European options by Monte Carlo with control variates and antithetic variance reduction. Calibrated on a real implied volatility surface and compared delta-hedging P&L to Black-Scholes and Heston. Technologies: Python, C++, SciPy, QuantLib, Stochastic Calculus, Monte Carlo, Fractional Brownian Motion, Calibration, Implied Volatility, Greeks	
Portfolio Risk Engine: Extreme Value Theory, Copulas and Expected Shortfall	Dec 2025 – Jan 2026
- Modelled marginal tail risk with GARCH and Generalized Pareto distributions (Peaks-Over-Threshold). - Modelled dependence with Student-t, Clayton and Gumbel copulas, and estimated tail dependence. - Computed VaR and Expected Shortfall by historical, parametric and copula Monte Carlo, with Ledoit-Wolf covariance. Backtested models with Kupiec, Christoffersen and Acerbi-Székely tests to quantify model risk. Technologies: Python, NumPy, SciPy, Statsmodels, arch, Scikit-learn, EVT, Copulas, GARCH, VaR, Expected Shortfall, Backtesting	

SKILLS

- Languages and tools:** Python, C++, SQL, Linux, Git, Docker, Kubernetes
- Libraries:** QuantLib, Statsmodels, CVXPY, NumPy, SciPy, Pandas, Matplotlib, Seaborn, PyTorch, TensorFlow, Keras, Scikit-learn
- Quantitative Finance:** Stochastic Calculus, Monte Carlo, Derivative Pricing, Greeks, VaR, Expected Shortfall, Calibration, Implied Volatility
- Probability & Statistics:** Stochastic Processes, Statistical Inference, Hypothesis Testing, Time Series, Regression, PCA, Covariance Estimation